

Bernardo Micol Righi

Back-End Developer | AI Systems | Tech Influencer

E-mail: bernardomicolrighi@outlook.com | Phone: +55 (51) 99601-1501 | Location: Rio Grande do Sul, Brazil

Site: righi.dev | GitHub: github.com/righibe | LinkedIn: linkedin.com/in/bernardo-righi | Lattes: lattes.cnpq.br/8608406696939587

PROFESSIONAL SUMMARY

Back-end developer with 2 years of experience in Python, AI, and automation. Researcher at Unisinos in partnership with Dell, focused on LLM agent observability using LangChain and OpenTelemetry. Cofounded Servidor dos Programadores, one of the six largest dev communities in Brazil on Discord (18k+ members). I also create content about programming and AI as a tech influencer. Fluent in English.

PROFESSIONAL EXPERIENCE

Researcher — AI Agents & Observability

Unisinos / Dell Technologies

Jun 2026 - Present · Remote

- Researching observability patterns for multi-agent systems, studying how to trace behavior, latency, and failures of AI agents in production.
- Investigating strategies to trace tool calls, agent handoffs, and context propagation — project in a research and architecture-prototyping phase alongside the Dell team.
- Delivered a comparative study of 6 embedding-clustering pipelines (K-Means, UMAP, PaCMAP, LocalMAP, densMAP + HDBSCAN) on real support-ticket data, with statistical validation (trustworthiness, DBCV) grounded in the scientific literature, informing the team's approach to automatic topic discovery.

Cofounder & Back-End Developer

Servidor dos Programadores

Jun 2022 - Present · Remote

- The community was growing fast but losing members due to poor onboarding and moderation — I built Python and Discord.py bots that automated welcomes and technical support, improving retention and reducing the team's manual work.
- As growth accelerated, the infrastructure needed zero-downtime deploys — I set up CI/CD pipelines with GitHub Actions that automatically test and ship updates on every push, eliminating manual deploys.
- Led the organization of technical events with sponsors like Alura, Shard Cloud, and Hostinger, coordinating speakers and live sessions for 18k+ developers.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, SQL

AI & Agents: LangChain, LangGraph, LCEL, RAG, LLMs, scikit-learn

Observability: OpenTelemetry, Jaeger, Grafana, Agent Tracing

DevOps & Infra: Docker, GitHub Actions, CI/CD, Git, Linux

Frameworks: Spring Boot, Django, Flask, Discord.py

Databases: MySQL, JPA / Hibernate

EDUCATION

B.Sc. in Computer Science

Universidade do Vale do Rio dos Sinos (Unisinos)

Feb 2026 - Present · São Leopoldo, Brazil · Lattes: lattes.cnpq.br/8608406696939587

COURSES AND CERTIFICATIONS

Java Marathon (Java Virado no Jiraya) · Python Worlds 1, 2 and 3 · HTML5 and CSS3 (Curso em Vídeo) · DevOps and Python Immersion (Alura) · Design and Social Enterprise HDSE Leadership (Lehigh University)

LANGUAGES

Portuguese: native | English: fluent (C1) | Spanish: intermediate (B1)

PROJECTS

AI Technical Support Agent

Python · LLMs · OpenAI GPT-4o · RESTful APIs

github.com/righibe/technical-suport-Agent

- Situation: first-line technical support tickets needed faster triage without waiting on a human agent.
- Action: built an autonomous agent in Python powered by GPT-4o to interpret tickets and generate contextual resolutions through RESTful APIs.
- Result: automated first-level responses for common cases without human escalation.

CI/CD Pipeline in Python

Python · GitHub Actions · Docker · Flask

github.com/righibe/ci-cd-pipeline-python

- Situation: no automated testing or deployment process — every update was a production-bug risk.
- Action: built a complete pipeline with GitHub Actions and Docker, running automated tests on every push and deploying to the container only if everything passed.
- Result: eliminated manual deploys and gained the guarantee that only validated code reached the environment.

Generative AI Discord Bot

Python · Discord.py · Open-source LLM API

github.com/righibe/bot-discord-IAgenerativa

- Situation: the community wanted real-time AI assistance inside Discord without relying on paid APIs.
- Action: built a bot with Discord.py integrating an open-source generative AI API, handling multi-turn conversation context.
- Result: gave 18k+ community members free, real-time AI support directly inside the server.

Housing Price Prediction — São Paulo

Python · scikit-learn · Decision Tree · Random Forest

github.com/righibe/predict-prices-sp

- Situation: wanted to apply supervised ML to a real problem, moving beyond theory.
- Action: trained and compared Decision Tree and Random Forest with scikit-learn to predict housing prices in São Paulo, analyzing feature importance.
- Result: identified location and square footage as the dominant variables and solidified my hands-on foundation in regression.

CRUD REST API — Django & SQL

Python · Django · REST Framework · MySQL

github.com/righibe/CRUD-Python-Django

- Situation: needed hands-on practice with Django's ORM and REST patterns beyond Flask.
- Action: built a full CRUD REST API with Django connected to a SQL database, structuring validation and serialization.
- Result: solidified Django fundamentals and REST API design patterns transferable to production work.

People Registry REST API — Java

Java · Spring Boot · JPA / Hibernate · Spring Web

github.com/righibe/Cadastro-pessoas

- Situation: only had Python experience and needed to understand the Java ecosystem.
- Action: built a REST API from scratch with Spring Boot and JPA, following the Controller / Service / Repository flow with DTO and ORM.
- Result: a solid foundation in Java and clarity on the MVC pattern for enterprise projects.